Acids Bases & Salts

Acid → Sour
H^+ ions

- Mineral
  - HCl
  - H_2SO_4
  - HNO_3
  - H_2CO_3
- Organic
  - Plants & Animals
  - (Weak)
  - Acetic Acid
  - Malic Acid
  - Citric Acid.

Bases
Bitter
Soapy
Slippery

Aq Sol. → Replaceable H^+ ions

HCOOH
(Acetic) ~ Basicity - No. of Replaceable H^+ ions.

Indicators
- Colour (Natural) / Synthetic
- Smell → Olfactory

Chemical Prop.

$Zn + 2HCl \longrightarrow ZnCl_2 + H_2\uparrow$

$2NaOH + Zn \longrightarrow Na_2ZnO_2 + H_2$

$Na_2CO_3 + 2HCl \rightarrow 2NaCl + H_2O + CO_2$

Reaction to test CO_2

$Ca(OH)_2 \longrightarrow CaCO_3 + H_2O$

Milkiness

if CO_2 purantage is a lot, Milkiness goes away.

$CaCO_3 + H_2O + CO_2 \longrightarrow Ca(HCO_3)_2$

Acid + Base.

$H_2SO_4 + NaOH \longrightarrow NaSO_4 + H_2O$

Neutralisation

- Phenopthalein $\rightarrow$ + HCl
- NaOH $\rightarrow$ Pink

Colourlus

Acid + Metal Oxides

→ Salt + H_2O

$Cu + HCl \rightarrow CuCl_2 + H_2O$

- Common Factor between Acids and Bases.

  ↳ Replaceable H^+ ions

$C_6H_{12}O_6$
$C_6H_{22}O_{11}$
C_2H_5OH

any compound which dissociate in H_2O to give H^+ ions is an acid

- Bases on Dissociation give OH^- ions.

Glucose → $C_6H_{12}O_6$

Alcohol → C_2H_2OH

Activity

- in case of HCl & H_2SO_4, bulb will glow.
  → Conduct electricity
- Alcohol & Glucose can't conduct electricity

Activity 2.2.7

$HCl \longrightarrow$ ✓
$C_6H_{12}O_6 \longrightarrow$ ✗

$HCl + H_2O \longrightarrow \underbrace{H_3O}_{\text{Hydronium}} + Cl^-$

$HCl(g) \longrightarrow H^+ + Cl^-$

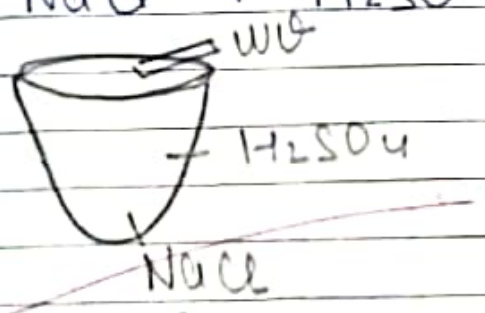
$H^+ + H_2O \longrightarrow H_3O^+$

Base $\longrightarrow OH^-$

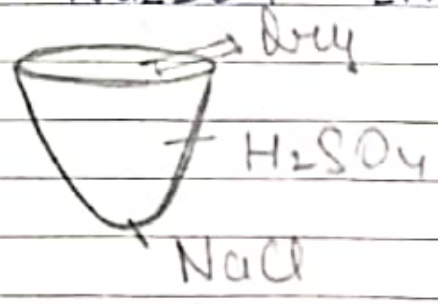
All Alkalis are bases but not vice versa.

Activity – 2.4

$NaCl + H_2SO_4 \longrightarrow Na_2SO_4 + 2HCl$



Tube A
Colour Changes



Tube B
No change

∴ Litmus needs to be wet for results.

- Base + H_2O

→ liberate OH^- ions

eg. $NaOH + H_2O \rightarrow Na + OH^-$

$KOH \xrightarrow{H_2O} K^+ + OH^-$

$MgOH \xrightarrow{H_2O} Mg^{2+} + 2OH$

Ph Scle

0 - 7 → Acidic
7 - 14 → Baisic

Acid → intuaRed
WeaK" → orange

weakBase → Light green

Base → Blue.

Plants → 7.0 → 7.8 ph

Acid Rain → 5-6 ph.

Digestive System — Stomach
HCl

Acidity → Antacid

ph ~~of soil~~ change
→ Cause Tooth Decay
- Bacteria
- Acid

→ Basic Tooth paste.
→ Enamel → Calcium

Honey Bee Stings
→ Acidic

Nettle Leaves
→ Methanoic Acid.
↗ Base
Dock plant

Some Imp. Compounds.

$2NaCl + 2H_2O \longrightarrow 2NaOH + Cl_2 +$

Bleaching powder

$Ca(OH)_2 + Cl_2 \longrightarrow CaOCl_2 + H_2O$

B.P.
↳ oxidising Agent

- make water germ free.

Baking Soda.
$NaHCO_3$ ↳ CO_2 evolved makes the cake soft & Spongy
↳ used as antacid
Soda Acid fire extinguisher

· Washing Soda.

$Na_2CO_3 \cdot 10H_2O$ —> Sodium Carbonate Decahydrate.

Soap, glass & paper industries.

Curing hardness of Water.

· More About Salts.

Acid + Base —> Salt + H_2O.

Acidic —> SA + H_2O
Neutral —> SA + H_2O ≈ 7

$HCl + NH_4OH \longrightarrow \underbrace{NH_4Cl_2}_{\text{Acidic}} + H_2O$

Chemicals from Common Salts.

NaOH —> Electrolysis of brine

NaCl + H_2O

Cathode (+)

Anode (−)

Electrolyte.

- Products of Chlor ALRali process.

→ Cl_2
→ H_2
→ NaOH

- Plaster of Paris (POP)

↳ Gypsum $\xrightarrow{373K}$ $CaSO_4 \cdot \frac{1}{2} H_2O$ (P

$CaSO_4 \cdot \frac{1}{2} H_2O + 1\frac{1}{2} H_2O \rightarrow$ Gypsum

$\frac{1}{2} H_2O \rightarrow$ One Molecule of H_2O shared by 2 units $CaSO_4$

POP → Calcium Sulphate Hemihydrate.